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研究简介 Research Profile

我的研究聚焦于如何将**人机智能协作（Human–AI Collaboration）**融入**可视分析（Visual Analytics）**，以支持复杂且具有领域特性的问题求解与决策。研究以数据可视化和人机交互为基础，结合人的专业知识与人工智能能力，设计面向领域专家的交互式分析流程，提升人工智能辅助分析的自主性、机器生成洞见的可解释性，以及智能决策的可信度。相关跨学科研究已获得多个领域研究人员与一线实践者的认可，并在实际工作场景中得到应用。

教育与科研经历 Experience

University of Konstanz — **Konstanz, Germany**
博士后研究员 — *2026.05 – 至今*
计算机与信息科学系；合作导师：Daniel A. Keim 教授；数据分析与可视化研究组成员

Monash University — **Melbourne, Australia**
研究学者 — *2025.01 – 2025.12*
信息技术学院；合作导师：Tim Dwyer 教授；Immersive Analytics Lab 成员

Singapore Management University — **Singapore**
博士 — *2022.01 – 2026.01*
计算机与信息系统学院；导师：Jiannan Li 教授、Yong Wang 教授

University of Electronic Science and Technology of China — **Chengdu, China**
学士 — *2015.09 – 2019.07*
计算机科学与工程学院；大数据研究中心成员

荣誉与奖励 Honors and Awards

2025: **Google 博士生奖研金** (Google PhD Fellowship)
研究方向：Advancing Explainable Quantum Computing through Visualization

2025: **IEEE VIS 2025 最佳论文荣誉提名奖** (IEEE VIS 2025 Best Paper Honorable Mention)
获奖论文：*"Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles"*

2024: **院长嘉奖名单** (SCIS Dean's List Award)
授予科研成果突出的前 3% 博士生

2024: **SMU 校长奖学金** (SMU Presidential Doctoral Fellowship)
授予持续取得卓越科研成果的前 10% 博士生

2023: **SMU 校长奖学金** (SMU Presidential Doctoral Fellowship)
授予持续取得卓越科研成果的前 10% 博士生

2019: **UESTC 计算机科学与工程学院卓越毕业生奖** (UESTC SCSE Outstanding Student Award)

学生指导与学术论文 Mentorship and Publications

— Mentored Work —
Qipeng Wang(mentee), **Shaolun Ruan**(mentor), Rui Sheng, Yong Wang, Min Zhu, Huamin Qu.
TrajLens: Visual Analytics for Constructing Cell Development Trajectories in Cross-Sample Analysis.
IEEE Transactions on Visualization & Computer Graphics (2025).

— Mentored Work —
Qipeng Wang(mentee), Rui Sheng, **Shaolun Ruan**(mentor), Xiaofu Jin, Chuhan Shi, Min Zhu.
SynthLens: Visual Analytics for Facilitating Multi-step Synthetic Route Design.
IEEE Transactions on Visualization & Computer Graphics (2025).

— Mentored Work —
Shaolun Ruan(mentor), Rohan Ramakrishnar(mentee), Chao Ren, Rudai Yan, Qiang Guan, Jiannan Li, Yong Wang.
Towards Explainable Quantum AI: Informing the Encoder Selection of Quantum Neural Networks via Visualization.
IEEE Transactions on Visualization & Computer Graphics (2026).

— Mentored Work —
Shaolun Ruan(mentor), Ribo Yuan(mentee), Qiang Guan, Yanna Lin, Ying Mao, Weiwen Jiang, Zhepeng Wang, Wei Xu, Yong Wang.
Venus: A Geometrical Representation for Quantum State Visualization.
Computer Graphics Forum (CGF) 42(3). https://doi.org/10.1111/cgf.14827

Shaolun Ruan, Rui Sheng, Xiaolin Wen, Jiachen Wang, Tianyi Zhang, Yong Wang, Tim Dwyer, Jiannan Li.
Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles.
IEEE Transactions on Visualization & Computer Graphics (2025).
IEEE VIS 2025 最佳论文荣誉提名奖（Best Paper Honorable Mention）

Shaolun Ruan, Qiang Guan, Paul Griffin, Ying Mao, Yong Wang.
QuantumEyes: Towards Better Interpretability of Quantum Circuits.
IEEE Transactions on Visualization & Computer Graphics (2023): 1–13.
https://doi.org/10.1109/TVCG.2023.3332999

Shaolun Ruan, Zhiding Liang, Qiang Guan, Paul Griffin, Xiaolin Wen, Yanna Lin, Yong Wang.
Violet: Visual Analytics for Explainable Quantum Neural Networks.
IEEE Transactions on Visualization & Computer Graphics (2023).
https://doi.org/10.1109/TVCG.2024.3388557

Shaolun Ruan, Yong Wang, Weiwen Jiang, Ying Mao, Qiang Guan.
Vacsen: A Visualization Approach for Noise Awareness in Quantum Computing.
IEEE Transactions on Visualization & Computer Graphics 29.01 (2022): 462–472.
https://doi.org/10.1109/TVCG.2022.3209455

Shaolun Ruan, Yong Wang, Hailong Jiang, Weijia Xu, Qiang Guan.
BatchLens: A Visualization Approach for Analyzing Batch Jobs in Cloud Systems.
Proceedings of DATE 2022. IEEE, 2022: 108–111.
https://ieeexplore.ieee.org/document/9774668

Manusha Karunathilaka*, **Shaolun Ruan***, Lin-Ping Yuan, Jiannan Li, Zhiding Liang, Kavinda Athapaththu, Qiang Guan, Yong Wang.
Intuit: Explain Quantum Computing Concepts via AR-based Analogy.
ACM Conference on Human Factors in Computing Systems (Late-Breaking Work) (2025).
https://dl.acm.org/doi/full/10.1145/3706599.3720085

Shaolun Ruan, Yong Wang, Qiang Guan.
Intercept Graph: An Interactive Radial Visualization for Comparison of State Changes.
2021 IEEE Visualization Conference (VIS). IEEE, 2021: 111–115.
https://doi.org/10.1109/VIS49827.2021.9623307

Hailong Jiang*, **Shaolun Ruan***, Bo Fang, Yong Wang, Qiang Guan.
Visilience: An Interactive Visualization Framework for Resilience Analysis using Control-Flow Graph.
Proceedings of IEEE PRDC 2023. https://ieeexplore.ieee.org/document/10356508

Xiaolin Wen, Tai Nguyen, **Shaolun Ruan**, Qiaomu Shen, Jun Sun, Feida Zhu, Yong Wang.
PonxziLens+: Visualizing Bytecode Actions for Smart Ponzi Scheme Identification.
IEEE Transactions on Visualization & Computer Graphics (2024).
https://ieeexplore.ieee.org/abstract/document/10794804

Rui Sheng, Zelin Zang, Jiachen Wang, Yan Luo, Zixin Chen, Yan Zhou, **Shaolun Ruan**, Huamin Qu.
CellScout: Visual Analytics for Mining Biomarkers in Cell State Discovery.
IEEE Transactions on Visualization & Computer Graphics (2025).
https://ieeexplore.ieee.org/abstract/document/11267149

Can Liu, Wenjie Jiang, **Shaolun Ruan**, Kotaro Hara, Yong Wang.
Accessible Fine-grained Data Representation via Spatial Audio.
IEEE Computer Graphics and Applications (2026). https://arxiv.org/abs/2604.08979

Feng Liang, Xiaolin Wen, Shixian Zhou, **Shaolun Ruan**, Yuxuan Du, Qiang Guan, Jun Ye, Yong Wang.
Visualization for Quantum Computing: Current State and Future Directions.
Under review by IEEE TVCG. [ResearchGate]

Peng Kuai, Yukun Yang, **Shaolun Ruan**, Junchi Xu, Yanjie Zhang, Lin Zhang, Min Zhu, Rui Sheng.
Within the MDT Room: Situated in Multidisciplinary Team-Grounded Agent Debate for Clinical Diagnosis.
arXiv preprint: https://arxiv.org/abs/2603.28393

Shuo Yan, Xiaolin Wen, **Shaolun Ruan**, Yanjie Zhang, Jiaming Mi, Yushi Sun, Huamin Qu, Rui Sheng.
InconLens: Interactive Visual Diagnosis of Behavioral Inconsistencies in LLM-based Agentic Systems.
arXiv preprint: https://arxiv.org/abs/2603.28106

Shaolun Ruan, Zhikang Wang, Furui Cheng, Rui Sheng, Kai Qian, Jiangning Song, Jiannan Li, Yong Wang.
ImaGene: A Multimodal Visualization Tool for Gene-powered Cancer Prognosis.
Under review.

受邀报告 Invited Talks

2026.01: **VIS Meets Quantum Computing**
ÉTS Quantum Seminar Series

2025.11: **Visualization Towards Addressing the Explanation of Scientific Problems**
RMIT University

2025.10: **VIS Meets Quantum Computing**
University of Toronto

2025.06: **Visualization Towards Addressing the Explanation of Scientific Problems**
University of Technology Sydney & Western Sydney University

2024.05: **VIS Meets Quantum Computing**
GAMES Webinar

2024.04: **VIS Meets Quantum Computing**
中南大学

2024.02: **Stepping Into the Era of Interpretable Quantum Computing**
University of Notre Dame — QuCS Lecture Series

2024.01: **Towards Making Your VIS Paper Writing Better**
电子科技大学

2024.01: **VIS Meets Quantum Computing**
四川大学

2023.12: **VAST Panel**
HKUST VisLab HAI Seminar

2023.11: **VIS Meets Quantum Computing**
HKUST

教学经历 Teaching

Studio 12, FIT3179 — Data Visualisation
Faculty of Information Technology, Monash University — *2025 Semester 2*
Student Lecturer

Studio 6, FIT3179 — Data Visualisation
Faculty of Information Technology, Monash University — *2025 Semester 2*
Student Lecturer

IS428 — Visual Analytics for Business Intelligence
School of Computing and Information Systems, SMU — *AY2023–24, Spring*
Teaching Assistant

CS711 — Reinforcement Learning
School of Computing and Information Systems, SMU — *AY2024–25, Fall*
Teaching Assistant

媒体报道 Media

- [SMU News] Google Announces 2025 PhD Fellowship Recipients. [报道]
- [QuCS Lecture] No. 59, QuCS Lecture Series. [文章]
- [SMU SCIS News] SCIS PhD Dean's List Award 2024. [报道]
- [SCU VIS News] Invited Talk at SCU VIS Lab. [中文报道]
- [GAMES Webinar] No. 324 — Talk and Panel Discussion. [中文报道]
- [UESTC News] The School of Computer Science Held the First "Outstanding Award" Ceremony. [中文报道]